BIODICAPT: Optimal Experimental Design

Loïc Lehnhoff

2026-09-01

Table of contents

Introduction

OED

Optimal Experimental Design (or OED for short) is a method for designing experiments that aims at maximising information gain while minimising a cost function. This cost function can encompass uncertainty, cost, time… it is up to the user to select the parameters to optimise.

The key concept is that **not all observations are equally informative**.

This concept[1] can be particularly interesting when working in participartory science, which is our case for the BIODICAPT project.

BIODICAPT

BIODICAPT is a French initiative that aims at monitoring biodiversity of agricultural lands on a large (*aka national*) scale through the use of various recording devices.

The project consists of two phases:

1. Data collection within a "research" network of agricultural plots.
2. Data collection within the 500 ENI network.

500 ENI is a network of farmers that can be recruited to work on BIODICAPT. However, their participation is *voluntary*, which raises some challenges. Therefore, the transition from the first phase to the second involves a significant shift in scale and philosophy for data collection, which is why OED will be needed.

Strategy

During the first phase of the project, data will be collected on ~125 agricultural plots. With our budget and logistical constraints, we aim at sampling 50 to 150 new plots within 500 ENI.

With OED, the question we want to answer is simple: **which points are most informative in the 500ENI network, given our previous data collection and our constraints?**

For that, we'll use explicative variables collected on site (temperatures, precipitation, type of land, …), response variables obtained from recorders (presence-absence of birds species mainly), and Species Distribution Models (SDM), in particular HMSC, which is a joint-SDM (JSDM) [1]. These models make use of several species distributions to find *latent* links between them, which supposedly improves the prediction accuracy.

They are opposed to single SDMs (Species Distribution Models applied to one species) and SSDMs (Stacked SDMs, are mutliple SMDs glued together, with no link between them).

HMSC

HMSC (Hierarchical Modelling of Species Communities) is a JSDM developped by Ovaskainen et al. (2017) [1], it is distributed as an open-source R-package on GitHub [2].

It is undoubtebly the most widely used JSDMin the last years. To summarise its principle, take a quick look at Figure 4, from their original publication [1]:

[1] Sometimes, also referred to as "adaptive sampling".

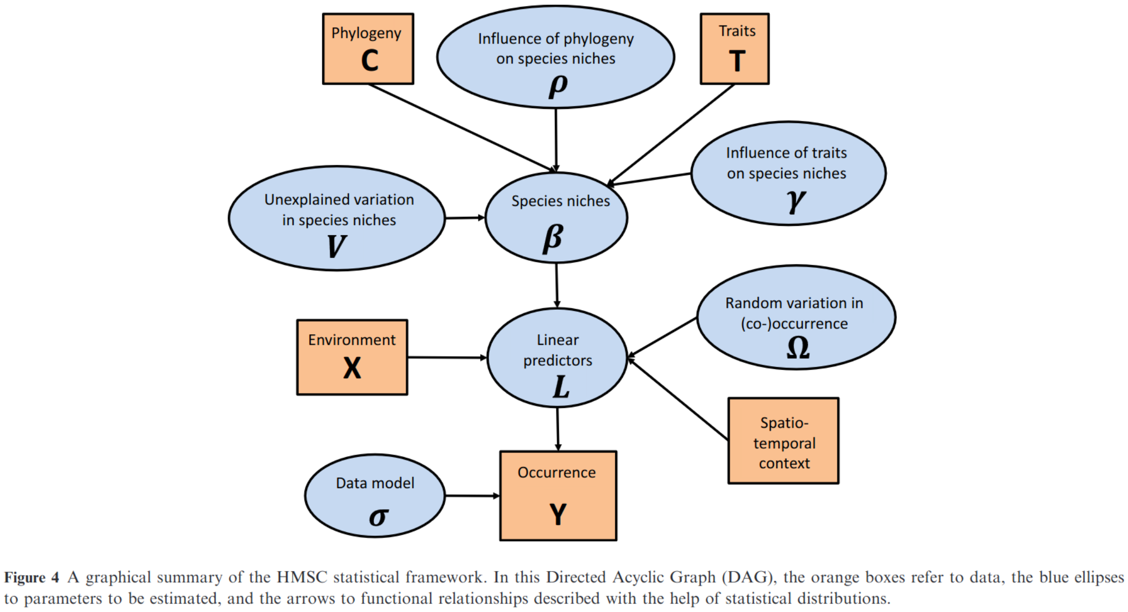


Figure 1: Copy of Figure 4 from Ovaskainen et al. (2017) [1]

GJAM

> Work in progress...
>
> Only HMSC has been implemented into our code for now. GJAM is under consideration, as well as other SDMs: RF, SSDM, deep-SDMs, ...
> This section will be updated when new results are available.

Proof-of-concept

Materials & Methods

The results from the first phase of the BIODICAPT data collection will not be available before early 2027. In the meantine, we can use another (but similar) dataset, already curated and known to work with HMSC: STOC [3].

The French Breeding Bird Survey (FBBS, or STOC in french) is a standardised multi-species abundance dataset, based on a citizen science program. It contains bird sightings and explanatory/environmental variables as the BIODICAPT dataset will. However, as it is already pre-processed, we can work directly with it, out of the box.

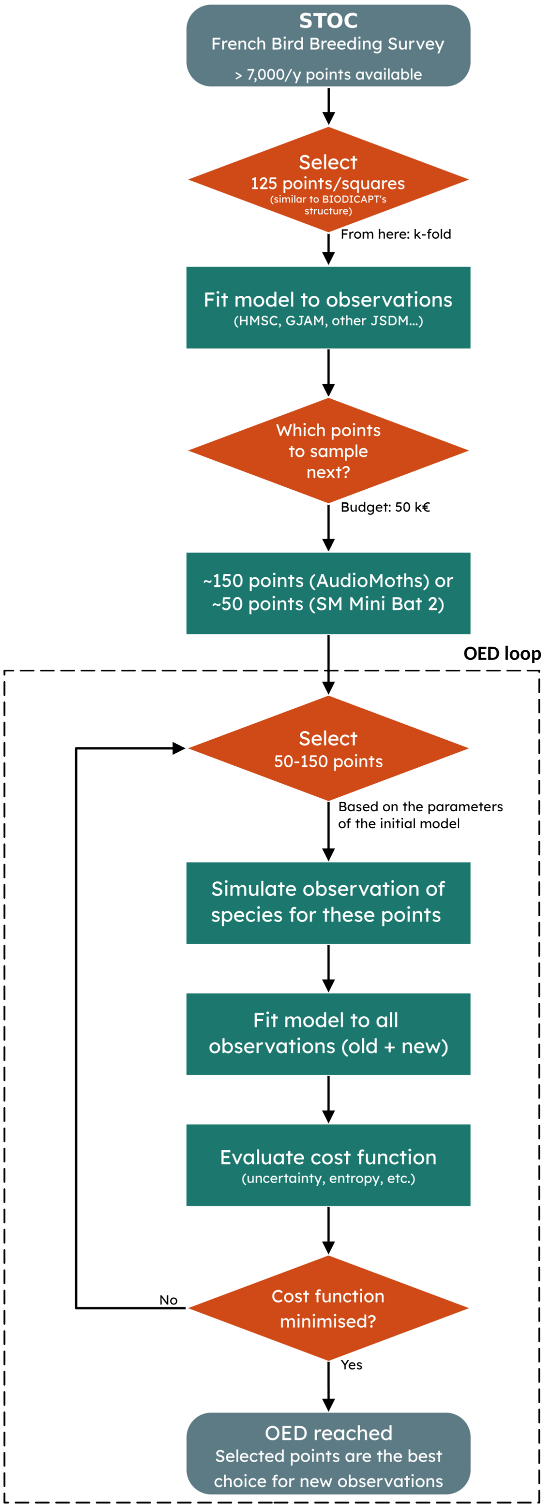


Figure 2: Flowchart of the proof of concept conducted with STOC dataset

Results compared

First, we want to make sure that, following the flowchart above, we can actually make better models by adding (50-150) new points to our dataset.

Reference results

To test this, we draw on *Mondain-Monval et al.* work [4]. However, it is only a base on which we build, not an exact replica, as our method is quite different from the original *Mondain-Monval et al.* publication:

Parameter	M.M.	Ours
Dataset	Simulated with `virtualspecies` package	STOC annual sampling
Number of species	50 species	15 species
Number of variables	10 (randomly selected from 33 available)	6 (direct from STOC)
Model type	Ensemble model (GLM + GAM + RF)	HMSC
Re-training strategy	Re-sampling of 1%, 10% or 50% of dataset	Addition of new samples (50)
Training size	2000 cells	125 cells + new data

With this context in mind, here are their results:

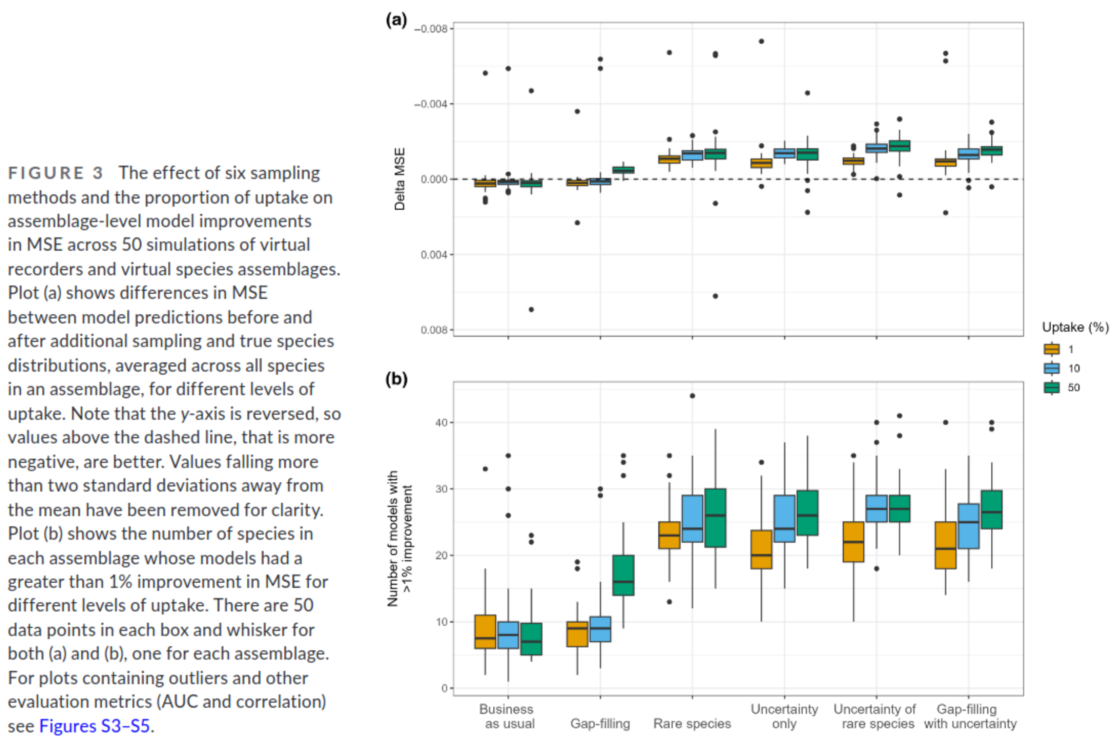
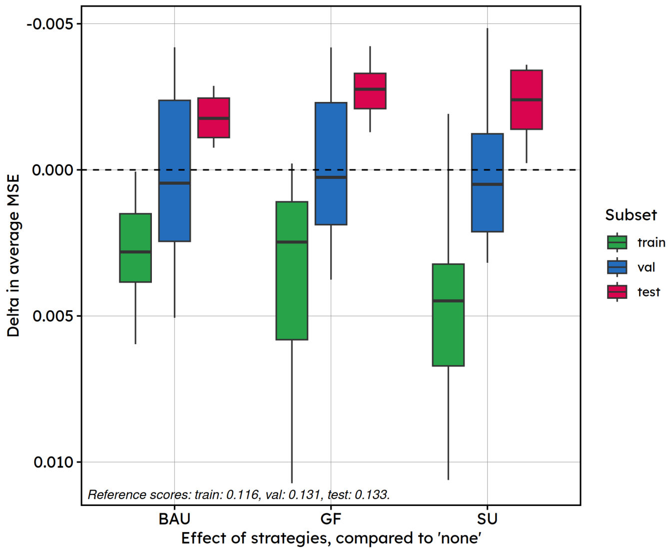


Figure 3: Copy of Figure 3a mondain-monval et al., 2024

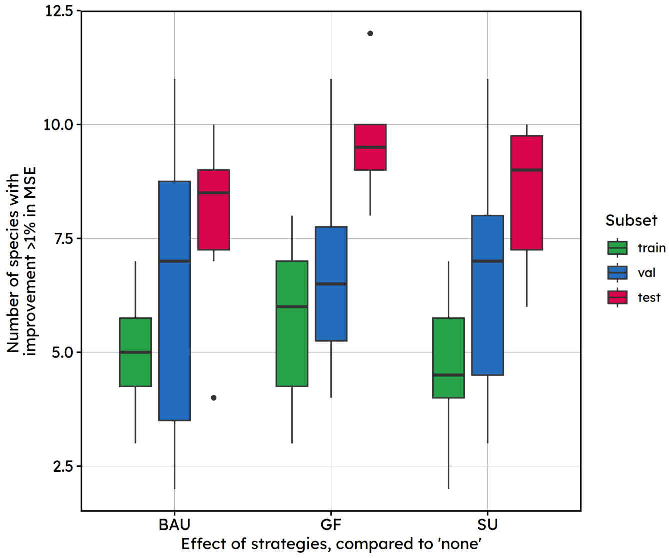
Our results

Boxplot differences



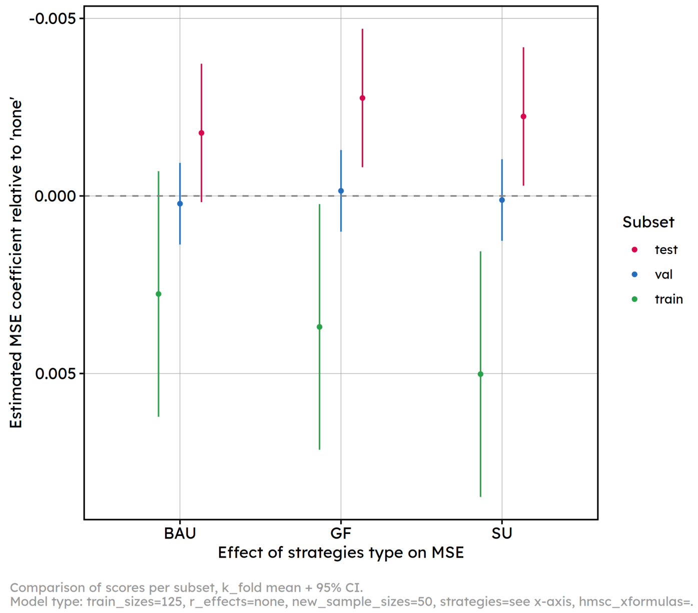
Distribution of per-fold differences (species-averaged within each fold, across k_fold).
Reference: train_sizes=125, r_effects=none, new_sample_sizes=50, strategies=none, hmsc_xformulas=~(NDVI + light_pollution + p_milieu + altit
Compared with: train_sizes=125, r_effects=none, new_sample_sizes=50, strategies=see x-axis, hmsc_xformulas=.

Boxplot improvement



Number of species per box: 15 .
Reference: train_sizes=125, r_effects=none, new_sample_sizes=50, strategies=none, hmsc_xformulas=~(NDVI + light_pollution + p_milieu + altit
Compared with: train_sizes=125, r_effects=none, new_sample_sizes=50, strategies=see x-axis, hmsc_xformulas=.

Estimated effects



BAU: Business-as-usual ; GF: Gap-filling ; SU: Simplified-uncertainty

Fitted model:

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: score ~ loop_element + (1 | k_fold) + (1 | species)
   Data: subset(scores_df, dataset == subset)

REML criterion at convergence: -3828.9

Scaled residuals:
    Min      1Q  Median      3Q     Max
-2.9440 -0.6839 -0.0940  0.6801  3.2178

Random effects:
 Groups   Name        Variance  Std.Dev.
 species  (Intercept) 5.805e-03 0.076191
 k_fold   (Intercept) 6.515e-06 0.002553
 Residual             7.390e-05 0.008596
Number of obs: 600, groups:  species, 15; k_fold, 10

Fixed effects:
                                     Estimate Std. Error        df t value
(Intercept)                         1.332e-01  1.970e-02 1.407e+01   6.759
loop_elementbusiness-as-usual      -1.774e-03  9.926e-04 5.730e+02  -1.788
loop_elementgap-filling            -2.759e-03  9.926e-04 5.730e+02  -2.779
loop_elementsimplified-uncertainty -2.238e-03  9.926e-04 5.730e+02  -2.255
                                   Pr(>|t|)
(Intercept)                        8.94e-06 ***
loop_elementbusiness-as-usual       0.07436 .
loop_elementgap-filling             0.00563 **
loop_elementsimplified-uncertainty  0.02452 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Conclusions: This model shows that, when adding 50 new points to our training subsets, test scores then vary depending on the strategy chosen to add those points. “GF” and “SU” show significantly lower MSE in test, while “BAU” shows no detectable effect.

Estimates are slightly better for “GF”, but remains comparable to “SU”.

All three effect sizes (0.0018-0.0028 in MSE units) are also small relative to the residual noise floor (SD 0.0086) and an order of magnitude smaller than between-species variability (SD 0.076), which meaningful for model tuning, but not a large practical effect at this sample size.

Comparison with Mondain-Monval et al.:

Mondain-Monval et al. [4] and we computed the difference in MSE between a base strategy (business-as-usual) and other strategies. However, our training sets were not built in the same way: - Mondain-Monval et al. [4] **changed a proportion of their training set** (1%, 10%; 50%) according to the chosen strategy. - We **added new samples to our training set** (50, with a base strategy of 125 samples) according to the chosen strategy.

Our figures also show scores in train/val/test. We assume Mondain-Monval et al. [4] show only scores in test.

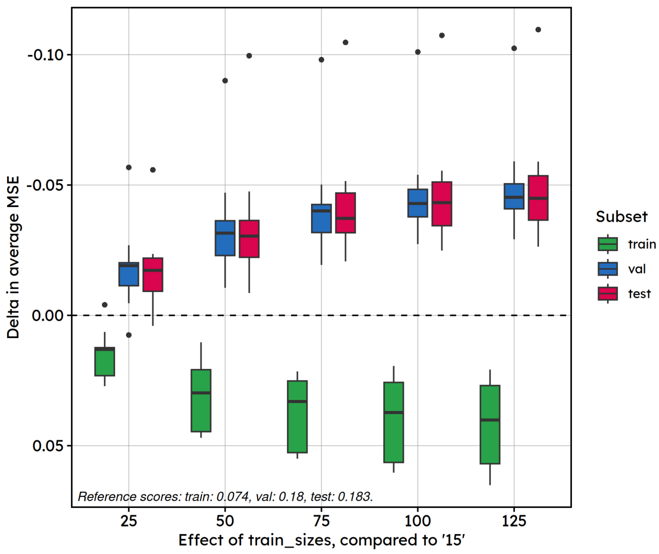
In our results, differences in MSE between resampling strategies are slight, less marked than those of [4]. Statistical tests reveal that there are significant differences in the test scores compared to no resampling. Interestingly, the “SU” strategy does not clearly yields to the best results (which was what we expected from [4]).

This is actually quite logical: the “SU” strategy adds training samples for which the predictions of the model are the most uncertain, which is why variability is wider in prediction.

Other results

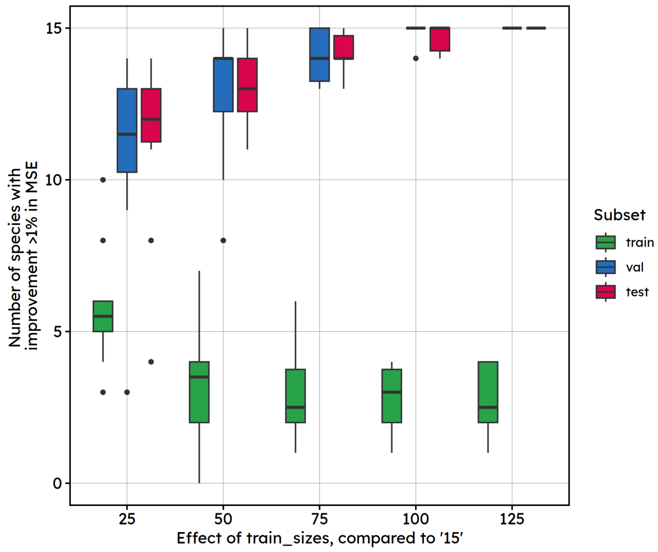
Training size

Boxplot differences



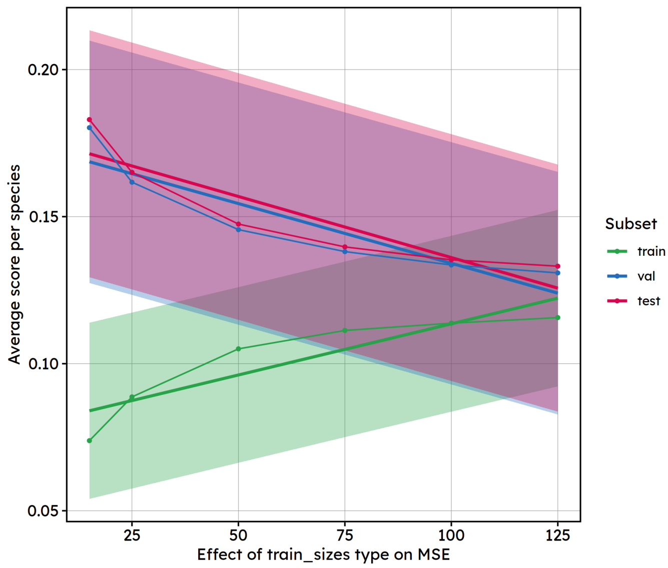
Distribution of per-fold differences (species-averaged within each fold, across k_fold).
Reference: train_sizes=15, r_effects=none, strategies=none, new_sample_sizes=0, hmsc_xformulas=~(NDVI + light_pollution + p_milieu + altitu
Compared with: train_sizes=see x-axis, r_effects=none, strategies=none, new_sample_sizes=0, hmsc_xformulas=.

Boxplot improvement



Number of species per box: 15 .
Reference: train_sizes=15, r_effects=none, strategies=none, new_sample_sizes=0, hmsc_xformulas=~(NDVI + light_pollution + p_milieu + altitu
Compared with: train_sizes=see x-axis, r_effects=none, strategies=none, new_sample_sizes=0, hmsc_xformulas=.

Estimated effects



Comparison of scores per subset, k_fold mean + 95% CI.
Model type: train_sizes=see x-axis, r_effects=none, strategies=none, new_sample_sizes=0, hmsc_xformulas=.

Fitted model:

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: score ~ loop_element + (1 | k_fold) + (1 | species)
   Data: subset(scores_df, dataset == subset)

REML criterion at convergence: -4051.2

Scaled residuals:
    Min      1Q  Median      3Q     Max
-2.4577 -0.6113 -0.0967  0.3963  7.4803

Random effects:
```

```
 Groups   Name        Variance  Std.Dev.
 species  (Intercept) 0.0067913 0.08241
 k_fold   (Intercept) 0.0000308 0.00555
 Residual             0.0005568 0.02360
Number of obs: 900, groups:  species, 15; k_fold, 10

Fixed effects:
               Estimate Std. Error         df t value Pr(>|t|)
(Intercept)   1.776e-01  2.140e-02  1.430e+01   8.298  7.7e-07 ***
loop_element -4.153e-04  2.003e-05  8.750e+02 -20.732  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

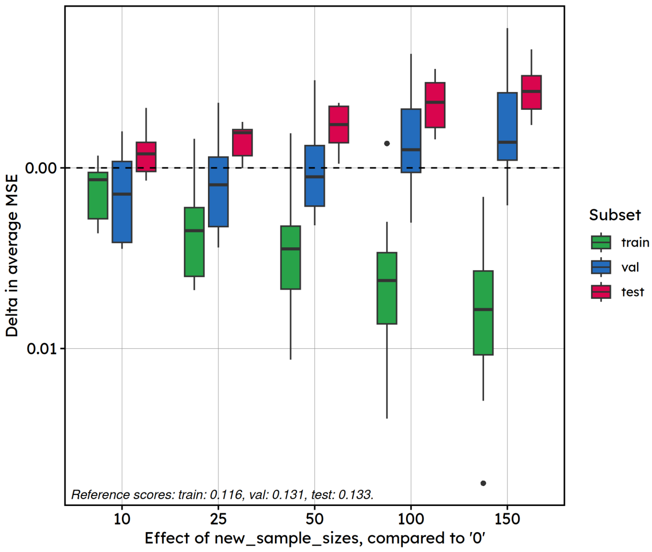
Conclusions:

Increasing training size produces a strong, highly significant (p «< 0.01), and meaningful reduction in test MSE.

Over 100 samples, this effect is comparable in magnitude to the variance explained by species groups.

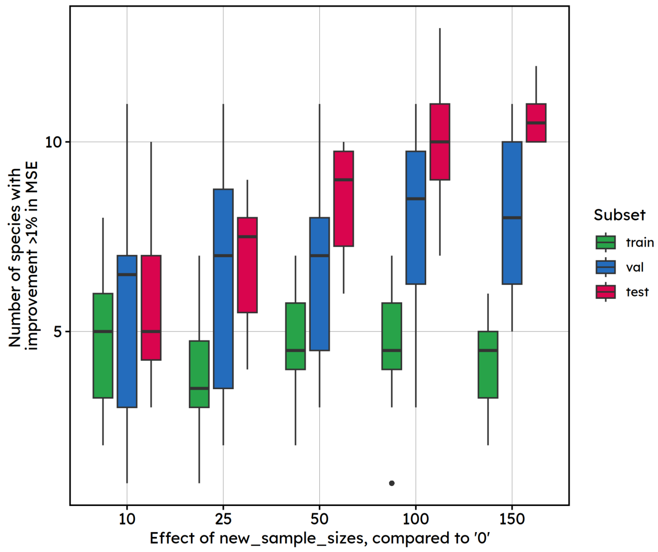
Number of new samples

Boxplot differences



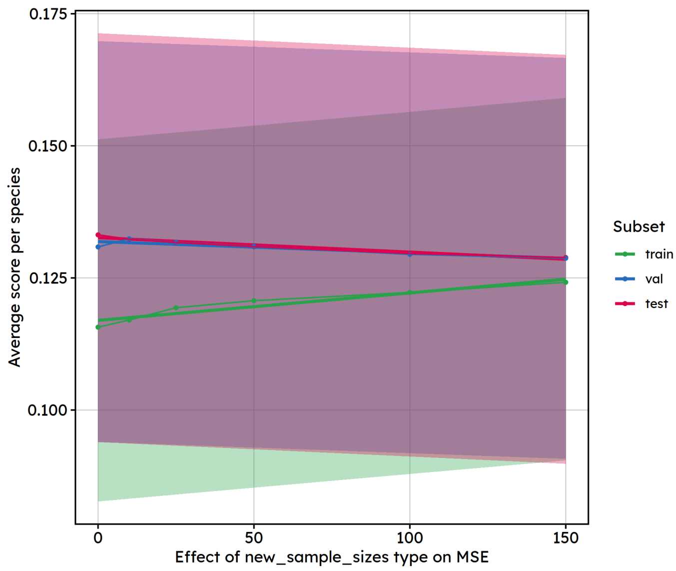
Distribution of per-fold differences (species-averaged within each fold, across k_fold).
Reference: train_sizes=125, r_effects=none, new_sample_sizes=0, strategies=simplified-uncertainty, hmsc_xformulas=~(NDVI + light_pollution
Compared with: train_sizes=125, r_effects=none, new_sample_sizes=see x-axis, strategies=simplified-uncertainty, hmsc_xformulas=.

Boxplot improvement



Number of species per box: 15 .
Reference: train_sizes=125, r_effects=none, new_sample_sizes=0, strategies=simplified-uncertainty, hmsc_xformulas=~(NDVI + light_pollution
Compared with: train_sizes=125, r_effects=none, new_sample_sizes=see x-axis, strategies=simplified-uncertainty, hmsc_xformulas=.

Estimated effects



Comparison of scores per subset, k_fold mean + 95% CI.
Model type: train_sizes=125, r_effects=none, new_sample_sizes=see x-axis, strategies=simplified-uncertainty, hmsc_xformulas=.

Fitted model:

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: score ~ loop_element + (1 | k_fold) + (1 | species)
   Data: subset(scores_df, dataset == subset)

REML criterion at convergence: -5858.7

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.1349 -0.6778 -0.0740  0.6209  3.3249

Random effects:
```

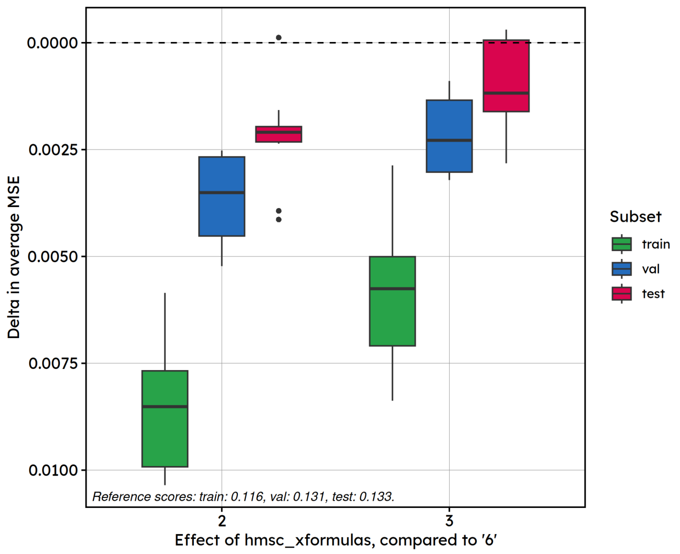
```
 Groups   Name        Variance  Std.Dev.
 species  (Intercept) 5.818e-03 0.076273
 k_fold   (Intercept) 5.929e-06 0.002435
 Residual             7.194e-05 0.008482
Number of obs: 900, groups:  species, 15; k_fold, 10

Fixed effects:
               Estimate Std. Error         df t value Pr(>|t|)
(Intercept)   1.326e-01  1.971e-02  1.405e+01   6.728 9.50e-06 ***
loop_element -2.727e-05  5.308e-06  8.750e+02  -5.137 3.44e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Conclusions: Similarly to training size, the effect is statistically significant (p «< 0.01) but small in intensity.

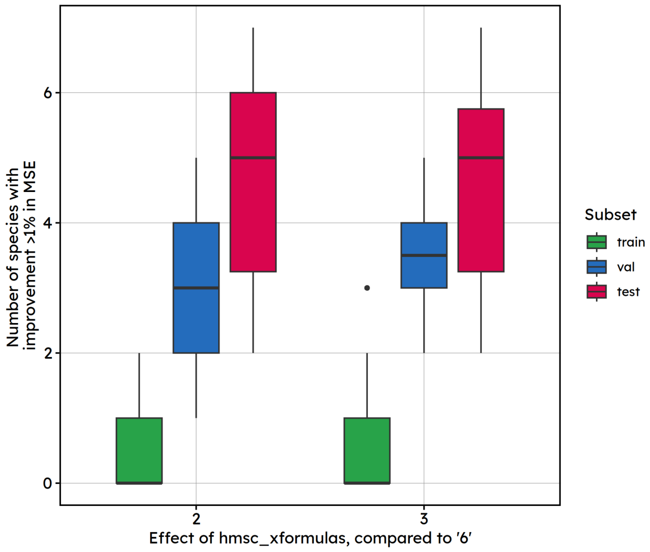
Number of explicative variables

Boxplot differences



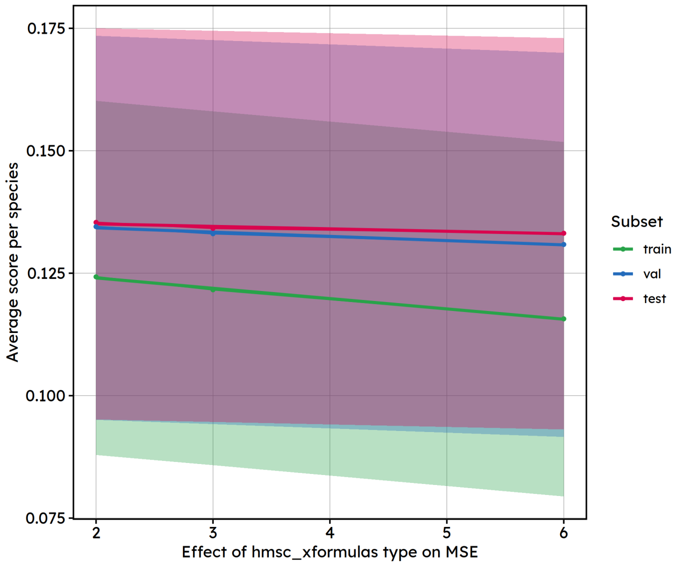
Distribution of per-fold differences (species-averaged within each fold, across k_fold).
Reference: train_sizes=125, r_effects=none, strategies=none, new_sample_sizes=0, hmsc_xformulas=~(NDVI + light_pollution + p_milieu + altitu
Compared with: train_sizes=125, r_effects=none, strategies=none, new_sample_sizes=0, hmsc_xformulas=see x-axi.

Boxplot improvement



Number of species per box: 15 .
Reference: train_sizes=125, r_effects=none, strategies=none, new_sample_sizes=0, hmsc_xformulas=~(NDVI + light_pollution + p_milieu + alti
Compared with: train_sizes=125, r_effects=none, strategies=none, new_sample_sizes=0, hmsc_xformulas=see x-axi.

Estimated effects



Comparison of scores per subset, k_fold mean + 95% CI.
Model type: train_sizes=125, r_effects=none, strategies=none, new_sample_sizes=0, hmsc_xformulas=see x-axi.

Fitted model:

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: score ~ loop_element + (1 | k_fold) + (1 | species)
   Data: subset(scores_df, dataset == subset)

REML criterion at convergence: -2798.6

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.69189 -0.72287 -0.07612  0.68723  3.02859

Random effects:
```

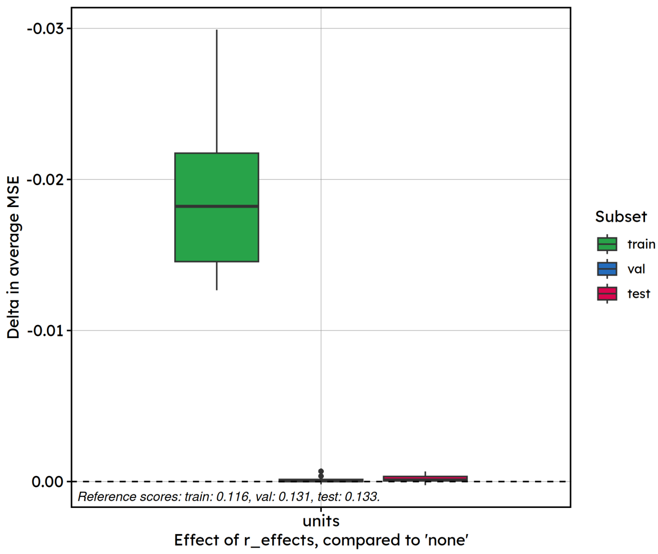
```
 Groups   Name        Variance  Std.Dev.
 species  (Intercept) 6.180e-03 0.078613
 k_fold   (Intercept) 6.188e-06 0.002488
 Residual             8.403e-05 0.009167
Number of obs: 450, groups:  species, 15; k_fold, 10

Fixed effects:
               Estimate Std. Error        df t value Pr(>|t|)
(Intercept)   1.361e-01  2.034e-02  1.410e+01   6.691 9.89e-06 ***
loop_element -5.045e-04  2.542e-04  4.250e+02  -1.984   0.0479 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Conclusions: There is weak evidence (p 0.05) that adding explicative variables reduces test MSE: there are only 3 points, which are not necessarily linearly distributed.

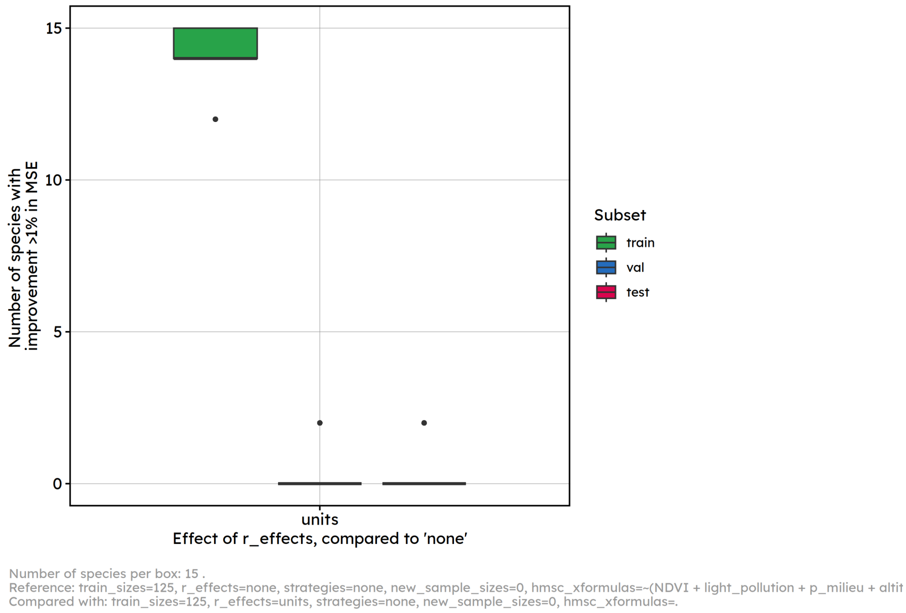
Random effects

Boxplot differences

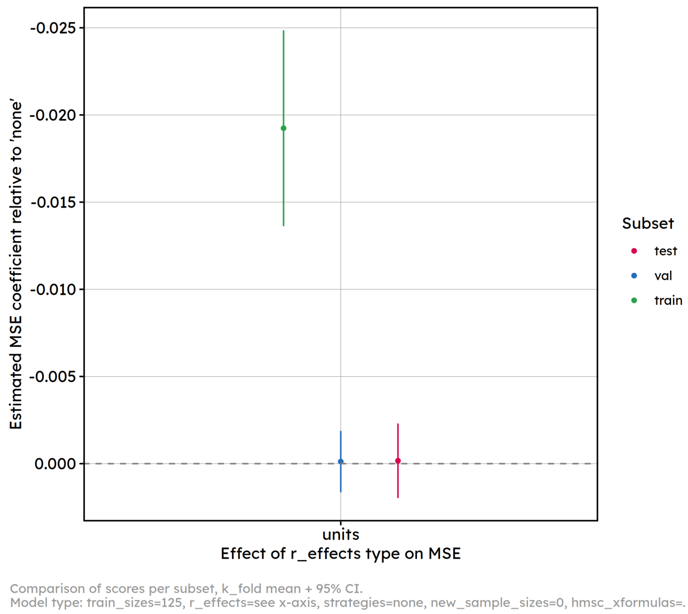


Distribution of per-fold differences (species-averaged within each fold, across k_fold).
Reference: train_sizes=125, r_effects=none, strategies=none, new_sample_sizes=0, hmsc_xformulas=~(NDVI + light_pollution + p_milieu + altit
Compared with: train_sizes=125, r_effects=units, strategies=none, new_sample_sizes=0, hmsc_xformulas=.

Boxplot improvement



Estimated effects



Fitted model:

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: score ~ loop_element + (1 | k_fold) + (1 | species)
   Data: subset(scores_df, dataset == subset)

REML criterion at convergence: -1814.1

Scaled residuals:
    Min      1Q  Median      3Q     Max
-2.7073 -0.6877 -0.1210  0.7215  2.6191

Random effects:
```

```
 Groups   Name        Variance  Std.Dev.
 species  (Intercept) 5.865e-03 0.076583
 k_fold   (Intercept) 6.392e-06 0.002528
 Residual             8.858e-05 0.009412
Number of obs: 300, groups:  species, 15; k_fold, 10

Fixed effects:
                    Estimate Std. Error        df t value Pr(>|t|)
(Intercept)        1.332e-01  1.980e-02  1.407e+01   6.723 9.49e-06 ***
loop_elementunits -1.658e-04  1.087e-03  2.750e+02  -0.153    0.879
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Conclusions: No significant difference in test MSE when the species random intercept is included vs. excluded ($p > 0.5$), despite a visible improvement in training fit.

This is a *textbook* signature of the random effect absorbing training-set noise without improving generalisation, which is expected between test units are very different, and not overlapping with training units. Surely a "distance" random effect would be more impactful.

Overall discussion

Five predictors were tested independently, one at a time, each in its own mixed model. To summarise the results:

- Training size has a positive effect on test performance. This is the strongest and most of this report .
- Sampling strategy: "GF" and "SU" perform significantly better than no resampling strategy.
- Number of new samples (independent of overall training size) and number of explicative variables both show statistically detectable but modest effects in practice.
- Explicit species random effects should consider distance between points instead of units, which has no impact.

Species random effect is high. As our aim is to generalise the predictions beyond the 15 species used here, this is fine, but it should be kept in mind that these results are highly species-dependant. K-folds random effects are negligible, which is expected and good thing, meaning that the partitioning of the dataset has little to no effect on scores.

Overall, increasing the number of training samples is the dominant lever for improving HMSC model performance, with returns flattening beyond roughly 100 sampled plots. However, this is only true for an initial random sampling of the STOC dataset, what if the initial sampled agricultural plots were not randomly chosen ? (i.e. in BIODICAPT, some explicative variables could be over-represented, which would skew the distributions). We'll have to change the dataset to see that!

1. Ovaskainen O, Tikhonov G, Norberg A, Blanchet FG, Duan L, Dunson D, et al. How to make more out of community data? A conceptual framework and its implementation as models and software. Ecology Letters [Internet]. 2017;20(5):561–576. Available from: https://doi.org/10.1111/ele.12757

2. Tikhonov G, Opedal ØH, Abrego N, Lehikoinen A, Jonge MMJ de, Oksanen J, et al. Joint species distribution modelling with the r-package Hmsc. Methods in Ecology and Evolution [Internet]. 2020 [cited 2020 Nov 23];11(3):442–447. Available from: https://besjournals.onlinelibrary.wiley.com/doi/abs/10.1111/2041-210X.13345

3. Vallé C, Poggiato G, Thuiller W, Jiguet F, Princé K, Le Viol I. Species associations in joint species distribution models: From missing variables to conditional predictions. Journal of Biogeography [Internet]. 2024 Feb [cited 2026 June 12];51(2):311–324. Available from: https://onlinelibrary.wiley.com/doi/10.1111/jbi.14752

4. Mondain-Monval T, Pocock M, Rolph S, August T, Wright E, Jarvis S. Adaptive sampling by citizen scientists improves species distribution model performance: A simulation study. Methods in Ecology and Evolution [Internet]. 2024 July [cited 2026 May 22];15(7):1206–1220. Available from: https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210X.14355